

Problem-Family Cognition

Core2 reconstructs what questions demand: cues, hidden state, first move, representation switch, wrong chain, checks, difficulty and transfer envelope.

This sample demonstrates:

- Surface wording is separated from problem family.
- First non-obvious move is usually axis -> $P=(x,0)/(0,y)$.
- Difficulty rises through hidden cues, inverse targets and discrimination.
- Core2 hands Core2A a transfer envelope, not a final workbook.

Sample topic

A point on an axis equidistant from two fixed points

Generated review fixture - not an official exam source

1. Cognitive anatomy

What does the question make the learner recognize and do?

SURFACE WORDING

axis/equidistance wording, diagram, equation or context

RECOGNITION CUE

axis constraint + two fixed points + equal distance

HIDDEN STATE

one coordinate fixed, one unknown

FIRST NON-OBVIOUS MOVE

write $P=(x,0)$ or $P=(0,y)$

REPRESENTATION SWITCH

geometry/context $\rightarrow$ squared-distance equation

WRONG CHAIN

midpoint shortcut used without structural justification

CHECK

verify $PA^2=PB^2$

2. Demand + transfer envelope

Safe variation changes the surface or target without importing new mathematics.

M0 -> M2

direct -> controlled variation -> representation transfer

M3 -> M4

inverse target -> hidden structure

M5 -> M6

method discrimination -> family discrimination

SAFE

axis change; unequal heights; verbal/diagram/equation shift; inverse coordinate; method choice

CONDITIONAL

contextual family discrimination requires model-selection authority

FORBIDDEN WITHOUT NEW AUTHORITY

3D distance, new locus theorems, or other mathematics not in the validated boundary